

Running Agri-Dosth on WhatsApp

The one authoritative guide to putting the grower's friend — for farmers and terrace/home gardeners alike — on WhatsApp: the official Cloud API for one-on-one chats, Baileys for group chats, the Meta Business setup, and how it all runs on our server.

-  Web chat — live
-  Webhook endpoint — live
-  Cloud 1:1 — awaiting tokens
-  Groups — awaiting spare number

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01 Is Agri-Dosth usable right now?

Yes. The web-chat version is live and healthy, powered by OpenAI:



Use it now → <http://187.127.166.193:8090> — growers type a question (or upload a plant photo) and Agri-Dosth answers.

Everything in this guide is about *adding* WhatsApp as a channel. None of that work touches the web chat — it keeps running exactly as today.

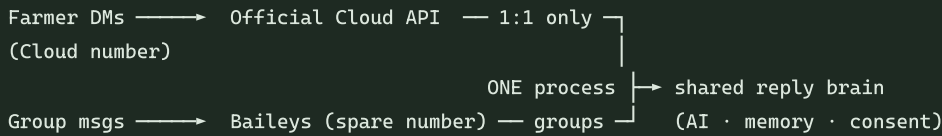
02 The big picture — two channels, not one

WhatsApp gives a bot two completely different ways to connect, and **one number cannot do both jobs**. This is the single most important thing to understand:

CHANNEL	ONE-ON-ONE (DMS)	GROUP CHATS
Official Cloud API — dedicated number, no ban risk	✓ Yes	✗ No — bots aren't allowed in normal groups
Baileys — spare real number, unofficial link	✓ Yes	✓ Yes — the only way to serve groups

So to reach farmers **everywhere** — DMs *and* village/co-op groups — we run a **hybrid**: both channels inside **one bot process**, sharing one database and one reply "brain" (so

consent, STOP, and data-erasure behave identically on both).



You do NOT need OpenClaw, Hermes, or Kapso. They use the same two transports we already have. More in section 8.

03 What numbers you actually need

Up to **two** numbers, depending on how far you go:

FOR 1:1 OFFICIAL CHATS

A Cloud API number

- **Now:** Meta's free **test number** (a +1 555... virtual number) — reaches only **5 pre-verified people**. Great for demos.
- **Later:** a **dedicated real number** + business verification, for real farmers.

FOR GROUP CHATS

A spare real SIM

- A cheap spare number you put in a phone, install WhatsApp on, and link by QR.
- This is the number that joins the farmer groups.



Your +1 555 640 7306 is the Cloud API test number. It is virtual — not a SIM, not on the WhatsApp app — so it works for **1:1 official chats only**. It **cannot** run Baileys or join groups. Groups need a separate, real spare number.



Registering a real number to the Cloud API removes it from the normal WhatsApp app and is largely irreversible. Dedicate a number you're happy to give up from the app.

04 Costs & the deadline that matters

The official API is **cheap-to-free for reactive replies** — but there's a clock:

ITEM	COST
Farmer-initiated replies within the 24-hour window	Free — until Oct 1, 2026
After Oct 1, 2026 — replies in-window	~₹0.115 per message
Marketing / utility templates	~₹0.86 / ~₹0.115 each
Baileys (groups)	No Meta message cost — carries ban risk instead

The old "1,000 free conversations/month" is gone (retired 2024). 18% GST applies in India. Going **direct to Meta** (not via a reseller) avoids per-message markups.

05 Part A — Official Cloud API (1:1 chats)

All of this happens at developers.facebook.com and business.facebook.com.

Step 1 — Business account & verification

At business.facebook.com, create a **Meta Business Account** (billing currency **INR**). Then Business Settings → Security Center → **Business Verification** — upload GST / MSME / utility bill. Lifts the 250-msg/day sandbox cap. *Takes days — start early.* (Not needed for the test number.)

Step 2 — Create the app & add WhatsApp

developers.facebook.com/apps → **Create App** → type **Business**. In the app, find **WhatsApp** → **Set up**. This creates your WhatsApp Business Account (WABA) and a test number.

Step 3 — Collect the 4 credentials

#	VALUE	WHERE	ENV VAR
①	Temporary access token	WhatsApp → API Setup (top)	WHATSAPP_CLOUD_TOKEN
②	Phone number ID	WhatsApp → API Setup (under "From")	WHATSAPP_PHONE_NUMBER_ID
③	App Secret	App Settings → Basic → Show	WHATSAPP_APP_SECRET
④	Verify token (you invent it)	any random string	WHATSAPP_VERIFY_TOKEN

⚠ **The temporary token (①) lasts 24 hours.** For longer, create a permanent **System User token** (Business Settings → System Users → assign the WhatsApp asset → Generate).

Step 4 — The webhook endpoint (already live)

Meta delivers incoming messages to an HTTPS URL — **already set up on our server** (nginx + auto-renewing cert):

```
https://agridosth.187-127-166-193.sslip.io/webhook/whatsapp
```

Step 5 — Put the credentials on the server

Add the four values to `/root/Agribot/.env` **directly on the box** — never paste tokens into chat or email:

```
ssh root@187.127.166.193
nano /root/Agribot/.env
# add all four:
WHATSAPP_CLOUD_TOKEN=<token @>
WHATSAPP_PHONE_NUMBER_ID=<id @>
WHATSAPP_APP_SECRET=<secret @>
WHATSAPP_VERIFY_TOKEN=<your string @>
```

All four must be present — the Cloud channel stays **off** unless every one is set.

Step 6 — Start the bot

```
cd /root/Agribot
pm2 restart agrifriend      # Baileys + Cloud webhook
pm2 logs agrifriend         # look for "Cloud API transport active"
```

Step 7 — Verify the webhook in Meta & subscribe

WhatsApp → **Configuration** → Webhook → **Edit**: Callback URL = the endpoint above;
Verify token = the same string you chose for ④; click **Verify and Save**, then **Subscribe**
to the **messages** field.

Step 8 — Add test recipients & test

WhatsApp → API Setup → Step 2 → "To" → **Manage phone number list** → add up to 5
numbers (each confirms with a code). Message the test number from one of those
phones — Agri-Dosth replies.

06 Part B — Baileys (group chats)

Baileys links to a **real spare WhatsApp number** like WhatsApp Web — the only way
Agri-Dosth can be in groups.

Step 1 — Pair the number

```
ssh root@187.127.166.193
cd /root/Agribot && pm2 restart agrifriend
pm2 logs agrifriend      # a QR code prints in the logs
```

On the spare phone: **WhatsApp** → **Settings** → **Linked Devices** → **Link a Device** → scan
the QR. Once linked, it stays paired (auth saved in **auth_info/**).

Step 2 — How groups work

- Add the spare number to the farmer group(s).
- The bot **only replies when triggered** — a message must contain the trigger word (**agrifriend** by default) or **@**-mention it.



Clean split: once 1:1 has moved to the Cloud number, set **BAILEYS_GROUPS_ONLY=true** — Baileys then serves **only** groups and ignores DMs, so the two channels never double-answer.

⚠ **Ban hygiene (built in):** reply-only, rate-limited, dedup'd. Never use Baileys for bulk/broadcast — that's what gets numbers banned.

07 Deployment & operations

Server `root@187.127.166.193` · repo `/root/Agribot` · branch `hardening/p0-p1` ·
Node 20 · PM2.

PM2 APP	WHAT IT IS	PORT
<code>agrifriend</code>	The WhatsApp bot — Baileys (groups) + Cloud webhook (1:1)	8080
<code>agridosth-web</code>	The web-chat page (separate)	8090
<code>agrifriend-ops</code>	Optional self-heal monitor	—

Deploy a new version

```
cd /root/Agribot
git checkout hardening/p0-p1 && git pull
npm run build           # src → dist
pm2 restart agrifriend  # and/or agridosth-web
```

08 Stability — how we compare to OpenClaw / Hermes

The worry was that those frameworks are "much more stable." The honest, evidence-based answer: we're at parity.

They use the same two transports we do — Baileys/whatsapp-web.js for the unofficial path, and the official Meta Cloud API for the stable path. There's no secret third method. Stability is the *engineering around* the transport — and our Baileys layer already implements the production checklist:

PRODUCTION-STABILITY FEATURE	OURS
Official Cloud API (no sessions, no ban risk)	✓ this build
Auth/session persistence	✓
Exponential-backoff auto-reconnect	✓
Correct logout-vs-transient handling	✓
Listener/socket-leak fix across reconnects	✓
Duplicate-delivery dedup (persisted)	✓
Rate-limiting + reply-only (anti-ban)	✓
Database-backed auth store (Redis/Postgres)	🔴 optional upgrade

Verdict: keep our codebase. The only enhancement bigger frameworks add is a database-backed auth store — nice for scale, not required for the pilot.

09 Current status & remaining checklist

Done **shipped**

- ✓ Cloud API adapter built — webhook, signature check, Graph API client, shared reply core
- ✓ Codex-reviewed · 220 tests passing · ~86% coverage
- ✓ Pushed to finalertserats-prog/Agribot and deployed on the server
- ✓ Web chat restarted on the new build — no regression
- ✓ HTTPS webhook endpoint live (cert auto-renews)

To run 1:1 on the official number **your move**

- Create/verify the Meta Business account
- Create the app + WhatsApp; get creds ①②③, pick ④
- Put the 4 vars in the server .env
- Restart the bot · Verify & Save in Meta · subscribe messages
- Add ≤5 test recipients and test

To run in group chats **your move**

- Get a real spare WhatsApp number (SIM) — the 555 test number can't do this
- Pair it via QR · add it to the farmer groups · trigger with agrifriend

10 Troubleshooting

SYMPTOM	LIKELY CAUSE / FIX
Webhook URL returns 502	The <code>agrifriend</code> process isn't running → <code>pm2 restart agrifriend</code>
Meta " Verify and Save " fails	Bot not running, or verify token in <code>.env</code> ≠ the one typed in Meta. They must match; restart after editing.
No reply to a real farmer on the test number	Test number only messages the 5 verified recipients. Add them, or move to a production number.
Cloud channel won't activate	All four <code>WHATSAPP_*</code> vars must be set. Check logs for "running Baileys-only".
Token stopped working after a day	Temporary token expired (24h) → switch to a permanent System User token.
Baileys keeps showing a QR	The spare number got unlinked → re-scan (Part B).

Maintained alongside the code in docs/WHATSAPP-SETUP-GUIDE.md · Agri-Dosth · 2026-07-25